ARTIFICIAL INTELLIGENCE | AUTONOMY | GOVERNANCE

Harnessing Intelligence

A Governance Architecture from Models to Complex Autonomous Systems

Artificial intelligence is moving from isolated model capability toward architectures that combine tools, memory, institutional knowledge, autonomous authority, cybersecurity, strategic interaction, perception, judgment, optimization, and action. The central challenge is no longer simply how to govern a model. It is how to build a harness at the same level at which intelligence becomes operational.

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Executive Brief

The contemporary conversation about artificial intelligence still tends to focus on models. Which model reasons better, which benchmark is higher, which model is safer, and which provider is winning. That framing is becoming increasingly incomplete. As artificial intelligence moves from conversational generation into tool use, institutional knowledge, persistent memory, autonomous action, inter-agent communication, and complex adaptive systems, the relevant unit of analysis becomes the full architecture in which intelligence operates.

This paper develops *harnessing* as the concept that connects that complete progression. In engineering practice, the harness increasingly refers to the runtime scaffolding around a model: the context, memory, tools, state, approval mechanisms, orchestration, verification, and persistence that allow a foundation model to behave as an agent. This paper accepts that technical meaning but extends it. For institutions, the harness must also encompass identity, provenance, knowledge governance, delegated authority, cybersecurity, behavioral monitoring, intervention, accountability, and eventually the collective behavior of interacting autonomous systems.

Central proposition
As the locus of intelligence expands, the locus of harnessing must expand with it. Harnessing must occur at the same level at which intelligence becomes operational.

The research program is organized as a seven-stage ladder. It begins with governed models and the recognition that artificial intelligence is broader than LLMs. It then moves through agentic architectures, institutional knowledge, autonomous agency, cybersecurity, spontaneous or emergent collaboration, and finally complex autonomous systems. The capstone is not a larger chatbot. It is an integrated architecture in which multiple forms of artificial intelligence participate in perception, judgment, optimization, and action.

The framework is cumulative. Earlier governance requirements do not disappear when a system becomes more sophisticated. Provenance remains necessary after autonomy is introduced. Knowledge governance remains necessary after tools and memory are added. Identity remains necessary when agents collaborate. Cybersecurity becomes more important as the architecture gains authority. The harness therefore expands as a nested governance structure:

$$H_1 \subset H_2 \subset H_3 \subset H_4 \subset H_5 \subset H_6 \subset H_7,$$

with

$$\boxed{H_n = H_{n-1} + \Delta H_n.}$$

The seven-stage ladder
Governed Models → Agentic Architectures → Institutional Knowledge → Autonomous Agency → Cybersecurity → Emergent Collaboration → Complex Autonomous Systems.

This paper also positions the argument within an emerging field of harness engineering. Microsoft now describes the agent harness as runtime scaffolding that turns a language model into an agent capable of performing sustained work. OpenAI has explicitly used the term *harness engineering* to describe the design of environments, abstractions, tools, feedback loops, and enforceable architectural constraints around agents. Recent research formalizes capability as emerging from a *model–harness–environment system*, while other work argues that future progress in agentic AI depends as much on scaling the harness as on scaling the underlying model. The contribution developed here is to extend this emerging engineering concept into an institutional theory of governable intelligence.

Abstract

The institutional debate surrounding artificial intelligence has developed in stages. The first challenge was to understand large language models and establish appropriate rules for their use. This quickly produced questions concerning prompting, provenance, validation, human accountability, and the protection of institutional information. The next transition occurred when language models ceased to operate as isolated conversational systems and became components of agentic architectures equipped with tools, memory, retrieval, and orchestration. A further transition connected these systems to persistent institutional knowledge through structured repositories, semantic architectures, and knowledge graphs. Autonomous agency then changed the problem again by allowing artificial systems to sustain objectives and actions over time. Cybersecurity consequently expanded from protecting information systems against conventional attacks toward protecting intelligent architectures capable of reasoning, adapting, and acting.

Research into spontaneous collaboration among autonomous agents introduces an additional systemic problem: individually governed agents may interact in ways that generate collective strategies, coalitions, or equilibria that were not explicitly designed. Finally, complex autonomous systems such as self-driving vehicles reveal the broader destination of this progression. Artificial intelligence becomes an architecture combining heterogeneous models, knowledge, context, perception, judgment, prediction, optimization, control, and action.

These subjects can appear to constitute separate research agendas. This paper argues that they are better understood as successive manifestations of a single architectural problem: **harnessing intelligence**. Harnessing is defined here not merely as the software infrastructure surrounding an AI agent, but as the complete technical and institutional architecture through which machine intelligence is connected to information, knowledge, tools, authority, other agents, and the external environment while remaining identifiable, attributable, observable, reconstructable, bounded, interruptible, and accountable.

Under this broader interpretation, the object of harnessing changes as artificial intelligence becomes more sophisticated. We begin by harnessing outputs. We then harness processes, knowledge, trajectories, security-sensitive capabilities, interactions among agents, and ultimately integrated systems of perception, judgment, optimization, and action. The central thesis is that the governance perimeter must expand whenever the locus of operational intelligence expands. Harnessing is therefore not another rung in the AI ladder. It is the architecture that makes the ladder governable.

1. The Research Problem Is Larger Than the Model

A great deal of the contemporary debate surrounding artificial intelligence begins with the large language model. This is understandable. LLMs created the first broadly accessible interface through which large populations could interact directly with sophisticated machine intelligence. Prompting consequently became the visible representation of AI use: a human expresses an objective in natural language, a model interprets the request, and an apparently intelligent response emerges.

The simplicity of that interface can obscure the complexity of artificial intelligence itself. A large language model is not synonymous with artificial intelligence, and conversational generation is only one form of machine cognition. Modern AI includes classification, prediction, computer vision, reinforcement learning, probabilistic inference, search, planning, anomaly detection, optimization, control, graph-based reasoning, multimodal perception, and increasingly powerful reasoning models. Some systems are generative. Others are discriminative. Some interpret. Some predict. Some optimize. Some act. The most consequential architectures are likely to combine several of these capabilities rather than rely upon one universal model.

This observation has been a recurring theme in the author's broader work on AI education and institutional adoption. The pedagogical starting point is therefore not merely to teach people how to use a chatbot. It is to understand the expanding toolbox of artificial intelligence and the architectural choices through which different forms of machine intelligence can be assembled into systems. This is the first reason the present paper begins with models but does not end with them.

The distinction becomes decisive when we examine an autonomous vehicle. It would be conceptually inadequate to describe a self-driving automobile as a large language model connected to steering and braking. The vehicle must perceive its environment through sensors and computer vision, estimate its location, classify objects, predict the behavior of pedestrians and other vehicles, understand contextual information, construct possible trajectories, optimize among those trajectories under safety and physical constraints, and execute control actions continuously. A language or reasoning model may increasingly contribute to some portions of that architecture, but it is neither necessary nor sufficient to explain the complete intelligent system.

The same reasoning applies outside transportation. An autonomous financial system, industrial system, cybersecurity architecture, scientific laboratory, or institutional decision platform will likely combine multiple forms of artificial intelligence. The relevant question is therefore not simply which model an institution uses. The deeper question is how intelligence is architected, and once that question is asked, another follows immediately: how is that architecture governed?

> ***"The strategic unit of artificial intelligence is no longer the model. It is the governed system in which intelligence operates."***

The concept of harnessing provides the bridge between these questions.

2. From the Engineering Harness to the Institutional Harness

In software engineering, an agent harness can be understood as the infrastructure surrounding a model and enabling it to operate as part of a larger system. The harness may supply instructions, memory, retrieval, tools, state, APIs, execution loops, evaluators, permissions, resource budgets, logging, and stopping rules. A model that can generate text becomes considerably more consequential when a harness allows it to retrieve information, execute software, maintain memory, call external systems, and repeat actions.

The relationship can be written simply as

$$S = M \oplus H,$$

where M denotes the model and H the harness. Yet this expression immediately reveals something important. The behavior of the resulting system cannot be explained by model capability alone:

$$B \neq f(M).$$

Instead,

$$\boxed{B = f(M, H, E),}$$

where E represents the environment.

This has direct institutional consequences. Consider two organizations using exactly the same frontier model. The first provides the model with no persistent memory, no tools, no external communications, and no authority to execute actions. Every consequential output is reviewed by a human. The second organization connects the same model to institutional databases, communications systems, code execution, persistent memory, financial infrastructure, and other autonomous agents. It permits the model to pursue objectives through repeated interactions without continuous human approval. The underlying model is identical, but the effective artificial intelligence system is not. The difference resides largely in the harness.

For institutional purposes, however, the engineering definition remains too narrow. Organizations do not merely need software orchestration. They need to know which artificial actor performed an action, what information it used, where that information originated, which authority permitted the action, what other systems participated, what constraints applied, whether the action can be reversed, and how the complete process can be reconstructed afterward. The institutional harness therefore includes technical mechanisms and governance mechanisms simultaneously.

Definition
Harnessing is the architecture through which artificial intelligence is connected to information, knowledge, tools, authority, other intelligent systems, and the external environment while remaining identifiable, attributable, observable, reconstructable, bounded, interruptible, and accountable.

This definition transforms harnessing from a technical component into an organizing concept for AI governance.

3. Harnessing as an Emerging Field

The term *harness* is increasingly acquiring a specific meaning in the engineering of artificial-intelligence agents. Microsoft's Agent Framework describes an agent harness as runtime scaffolding that turns a language model into an agent capable of sustained work by managing model and tool calls, conversation state, context, approval policies, planning, memory, and multi-step execution. OpenAI has similarly used the term *harness engineering* to describe a shift in the work of engineering teams: humans increasingly design environments, specify intent, create tools, and build feedback loops that allow agents to perform reliable work rather than manually implementing every step of the system.

Anthropic's work on long-running and managed agents illustrates a related architectural transition. As models become more capable, infrastructure assumptions that were reasonable for earlier systems can themselves become constraints. Context management, durable sessions, execution environments, tool interfaces, and the separation of model reasoning from the environments in which actions are executed become first-class architectural choices. The harness is therefore not simply a wrapper. It increasingly mediates the relationship between the "brain" of an artificial agent and the "hands" through which that intelligence acts.

The emerging academic literature makes the same point more formally. Zhong and Zhu characterize autonomous software capability as a property of a *model–harness–environment system* and identify task specification, context selection, tool access, project memory, task state, observability, failure attribution, verification, permissions, auditing, and intervention as harness responsibilities. Gu similarly argues that the next major bottleneck in agentic AI is not merely model scaling but *system scaling*: the design of persistent, modular, auditable, and verifiable harness architectures around increasingly capable models. Hu Wei's empirical study of agent systems identifies recurring harness design dimensions in context management, tool systems, sub-agent architecture, safety mechanisms, and orchestration.

This literature is important because it makes explicit what institutional governance increasingly requires: capability is not a property of a model alone. It emerges from the relationship among a model, its harness, and its environment. The present paper accepts this engineering insight but extends it. Once artificial intelligence is connected to proprietary knowledge, institutional memory, delegated authority, communication channels, external infrastructure, and other autonomous agents, the harness must encompass more than runtime reliability. It must also encompass identity, provenance, authority, cybersecurity, accountability, intervention, and systemic behavior.

Engineering Harness

context + memory + tools + state + orchestration + verification

$\Downarrow$

Institutional Harness

engineering harness + identity + provenance + knowledge + authority + security + accountability

$\Downarrow$

Systemic Harness

institutional harness + interaction + emergence + system-level control

> ***"The engineering harness turns model capability into operational agency. The institutional harness must turn operational agency into governable agency."***

This extension is central to the paper's contribution. The engineering literature asks how a model can be surrounded by infrastructure that allows it to perform reliable work. The institutional question is larger: how must that architecture evolve when the artificial intelligence being harnessed progresses from generating an answer, to using tools, to activating institutional knowledge, to exercising autonomous authority, to interacting strategically with other agents, and finally to participating in a complex adaptive system?

4. The Central Hypothesis: Governance Must Follow Intelligence

The broader research program can be interpreted as a sequence of expansions in the locus of intelligence. At the beginning, intelligence is concentrated primarily in a model. The institutional problem is correspondingly narrow: govern inputs and outputs. Once tools and orchestration are introduced, intelligence becomes partly architectural, and governance must encompass the workflow. When institutional knowledge is connected to the architecture, cognition becomes dependent upon persistent information structures, so governance must encompass what the system knows and how it knows it.

When autonomy is introduced, intelligence operates across time and governance must encompass trajectories. When these systems operate in adversarial environments, cybersecurity must encompass the complete architecture. When multiple autonomous agents interact, intelligence becomes distributed across a network and governance must encompass relationships and collective behavior. Finally, in a complex autonomous system, intelligence is distributed across heterogeneous models, knowledge structures, prediction mechanisms, optimization engines, control systems, and feedback loops. Governance must therefore operate at the level of the integrated system.

Central hypothesis

$$\text{Expansion of Intelligence} \Longrightarrow \text{Expansion of Harnessing.}$$

The more the locus of intelligence expands beyond a model, the more the locus of governance must expand beyond model oversight.

The nesting of the harness is important because governance is cumulative:

$$H_1 \subset H_2 \subset H_3 \subset H_4 \subset H_5 \subset H_6 \subset H_7.$$

When we move from governed prompting to agentic systems, we do not stop caring about provenance. When institutional knowledge is introduced, tool permissions do not disappear. When autonomy is introduced, knowledge governance remains necessary. When agents begin interacting, identity remains essential. The more sophisticated architecture inherits the governance requirements of all preceding architectures.

Thus,

$$\boxed{H_n = H_{n-1} + \Delta H_n.}$$

The question at every stage becomes: what new form of capability has appeared, and what new dimension must therefore be added to the harness?

5. The Seven-Stage Ladder

VII. Complex Autonomous Systems

↑

VI. Spontaneous and Emergent Collaboration

↑

V. Cybersecurity

↑

IV. Autonomous Agency

↑

III. Institutional Knowledge

↑

II. Agentic Architectures

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I. Models, Prompting, and Governed AI

Table 1: The AI Ladder and the Progressive Expansion of Harnessing

Stage	Architecture	Principal Capability	Object of Harnessing
I	Models and governed AI	Generation, prediction, classification and reasoning	Inputs, outputs, identity and provenance
II	Agentic architectures	Tool use and orchestrated workflows	Processes, tools, memory and execution
III	Institutional knowledge	Persistent organizational context	Knowledge, evidence, memory and lineage
IV	Autonomous agency	Persistent decision and action	Authority, objectives and trajectories
V	Cybersecurity	Operation under adversarial exposure	Attack surface, containment and resilience
VI	Emergent collaboration	Strategic interaction among agents	Relationships, coalitions and collective behavior
VII	Complex autonomous systems	Integrated perception, judgment and optimization	System-level cognition and action

The ordering matters. The first stage establishes that artificial intelligence is broader than LLMs. The second explains how models become components inside organized workflows. The third introduces institutional knowledge, giving AI systems access to persistent and structured organizational context. The fourth introduces autonomous agency and therefore delegated authority over sequences of actions. The fifth considers the cybersecurity consequences of connecting these systems to institutional resources. The sixth recognizes that multiple autonomous agents may interact and generate collective behavior. The seventh brings the preceding elements together inside complex intelligent systems in which heterogeneous forms of artificial intelligence perform complementary cognitive and operational functions.

6. Stage I: Models, Prompting, and the First Governance Problem

The first stage of the research concerns institutional adoption and the governance of models. It begins with a pedagogical principle that becomes increasingly important as AI interfaces become easier to use: ease of use should not be confused with simplicity of the underlying technology. A user may interact with an LLM through natural language without understanding transformers, embeddings, probabilistic generation, optimization, or inference. Yet institutions making consequential decisions need a more sophisticated understanding of what kind of model is being used, what it can and cannot do, and how its outputs should enter professional processes.

The author's work on AI governance and education has repeatedly emphasized that the institutional challenge begins before autonomy. It begins with purpose, identity, accountability, evidence, and human responsibility. If an AI-assisted memorandum, valuation, legal analysis, scientific conclusion, or policy recommendation becomes institutional work product, the organization should be capable of understanding which information entered the system, which model processed it, what output was generated, and who accepted responsibility for its eventual use.

At this level the harness can be represented as

$$H_1 = \{I_d, M, P, D, V, R\},$$

where I_d represents identity, M the approved model, P prompting policy, D data governance, V validation, and R review and responsibility. The corresponding lineage is

$$\text{Source} \rightarrow \text{Context} \rightarrow \text{Prompt} \rightarrow \text{Model} \rightarrow \text{Output} \rightarrow \text{Human Judgment}.$$

This apparently elementary architecture establishes several principles that survive throughout the entire research ladder. Identity matters because the institution must know who or what produced an output. Provenance matters because the organization must understand where information originated. Validation matters because probabilistic generation cannot automatically be treated as institutional truth. Human authority matters because the system remains embedded in an institutional decision process.

$$\boxed{H_1 = \text{Harness of Governed Cognition.}}$$

7. Stage II: Agentic Architectures and the Transition from Output to Process

The second research stage begins when AI stops being principally a conversational interface and becomes an architecture. The LLM or reasoning model is connected to tools, memory, retrieval, APIs, specialized models, evaluators, and orchestration mechanisms. Instead of responding once to a prompt, the system may perform a sequence of operations in pursuit of a goal.

This transition is central because the intelligence of the system no longer resides entirely in the model. Part of it resides in the architecture. Consider an institutional research agent. It may interpret an assignment using a reasoning model, search an approved document repository, retrieve relevant evidence, invoke a quantitative model, generate alternative hypotheses, call another model to critique the analysis, revise its conclusion, and produce a final memorandum. No single model invocation explains the resulting capability. The architecture itself contributes intelligence by organizing models, information, and tools into a purposeful sequence.

The relevant structure becomes

$$\text{Objective} \rightarrow \text{Orchestration} \rightarrow \{M_1, T_1, M_2, T_2, \ldots\} \rightarrow \text{Result.}$$

Harnessing must therefore expand from the output to the process:

$$H_2 = H_1 + \{T, M_e, S, O, B\},$$

where T denotes tool permissions, M_e memory controls, S state management, O orchestration governance, and B computational or financial budgets.

The significance of this stage for provenance is substantial. It is no longer sufficient to know which source supported the final response. The institution may need to reconstruct the computational trajectory through which the result emerged:

$$O_0 \rightarrow A_1 \rightarrow O_1 \rightarrow A_2 \rightarrow O_2 \rightarrow \cdots \rightarrow A_n.$$

A mistake may have entered at one intermediate stage and propagated through all subsequent operations. A compromised tool may have returned incorrect information. A model may have created an assumption that later components treated as fact. The final output may therefore be impossible to understand without examining the process.

$$\boxed{H_2 = \text{Harness of Orchestrated Agency.}}$$

8. Stage III: Institutional Knowledge and the Governance of Machine Belief

The third stage concerns institutional knowledge. This is not simply another tool added to the agent. It represents a major transformation in the architecture because it changes the informational world within which artificial intelligence operates.

General-purpose models contain broad statistical representations derived from training, but institutions possess knowledge that is private, dynamic, relational, contextual, and frequently authoritative. Contracts, policies, transactions, customer histories, board decisions, internal research, risk limits, organizational structures, precedents, and institutional experience cannot simply be assumed to exist reliably within model parameters. An institution that wants durable AI capability therefore requires an architecture through which this knowledge can be activated while remaining governed.

Knowledge architectures may include document repositories, retrieval systems, vector databases, relational databases, ontologies, semantic layers, persistent memory, and knowledge graphs. The objective is not merely to give the model more information. It is to provide artificial intelligence with a structured representation of the institution's world.

$$\boxed{\text{General Intelligence} + \text{Institutional Knowledge} + \text{Context.}}$$

This stage connects directly to the author's work on second brains and knowledge graphs. A second brain is not simply a larger memory. It is an architecture for activating institutional knowledge. Which source is retrieved, which relationship is traversed, which precedent is treated as authoritative, which contradictory evidence is surfaced, and which assumptions are inherited all influence the resulting machine judgment. The architecture of navigation therefore becomes part of the architecture of reasoning.

The distinction between knowledge and belief becomes especially important. Suppose an AI system infers from several transactions that a counterparty may represent elevated risk. If that inference is stored in persistent memory and later retrieved without provenance, a future agent may interpret the machine-generated hypothesis as an established institutional fact. Repeated cycles can create epistemic contamination. The harness must therefore preserve lineage:

$$\text{Claim} \rightarrow \text{Inference} \rightarrow \text{Evidence} \rightarrow \text{Source.}$$

This is substantially more demanding than citation. The institution must distinguish among observed fact, retrieved claim, machine inference, hypothesis, and validated institutional knowledge:

$$\text{Observed Fact} \neq \text{Retrieved Claim} \neq \text{Machine Inference} \neq \text{Hypothesis} \neq \text{Validated Knowledge.}$$

Properly designed, a knowledge graph can therefore become part of the harness itself because it can preserve semantic identity, provenance, relationships, access control, and the distinction between source evidence and machine-generated inference.

"At the knowledge stage, provenance becomes epistemic governance."

$$\boxed{H_3 = \text{Harness of Institutional Cognition.}}$$

9. Stage IV: Autonomous Agency and the Governance of Trajectories

The transition from agentic architecture to autonomous agency represents one of the most important discontinuities in the ladder. An agentic system can use tools without being truly autonomous. A human may still define every consequential objective and approve every important action. Autonomy begins when the system acquires discretion over what happens next and can sustain a sequence of actions through changing states.

The relevant unit becomes

$$\tau = (s_0, a_0, s_1, a_1, \ldots, s_T).$$

The system observes state s_t, forms a judgment, selects action a_t, observes the resulting state s_{t+1}, and continues. The governance problem therefore moves from controlling a workflow to controlling a trajectory.

This is conceptually related to principal-agent theory, but the analogy is incomplete. Human organizational agents operate within legal, cultural, professional, and hierarchical systems. Artificial agents may operate at machine speed, replicate processes, communicate with other artificial actors, invoke tools dynamically, and maintain persistent operation without the same forms of social accountability. The harness must consequently encode institutional authority more explicitly.

$$H_4 = H_3 + \{G, P, A, L, E, X\},$$

where G represents goal constraints, P permissions, A delegated authority, L limits, E escalation, and X intervention or shutdown mechanisms.

The autonomy question
The central governance question changes from “What is this model authorized to say?” to “What is this system authorized to cause?”

Effective autonomy depends upon several interacting variables:

$$\mathcal{A} = f(I, K, T, P, M, \tau),$$

where I is intelligence, K knowledge, T tools, P permissions, M memory and persistence, and τ the capacity to sustain behavior through time. This formulation explains why focusing exclusively on frontier-model capability can misrepresent institutional risk. A highly capable model with no tools and no permissions may possess limited operational authority. A less capable model with persistent memory, institutional knowledge, extensive permissions, and autonomous execution may have considerably greater capacity to affect the institution.

The relevant governance object is therefore effective agency, not intelligence in isolation.

$$\boxed{H_4 = \text{Harness of Autonomous Authority.}}$$

10. Harnessing Authority: Identity Must Travel with Delegation

Autonomous systems make identity substantially more complicated. It is insufficient to record that "the AI" executed an action. A system may contain several models, multiple agents, dynamically created sub-agents, external tools, and delegated processes. The chain of authority might resemble

$$\text{Board} \to \text{Management} \to \text{AI System} \to A_1 \to A_2 \to T \to \text{Action.}$$

The central governance requirement is continuity. If A_1 delegates a task to A_2, and A_2 invokes tool T, the institutional authority and constraints governing A_1 should not simply disappear. Identity, permissions, provenance, limits, and accountability should travel with delegation.

Delegated intelligence must carry delegated governance.

The analogy to financial authority is useful. An employee authorized to spend a limited amount cannot manufacture unlimited authority merely by delegating the purchase to another employee. In artificial systems, however, delegation can occur dynamically and at machine speed. Harnessing must therefore make the inheritance of authority computationally explicit.

11. Stage V: Cybersecurity and the Expansion of the Attack Surface

The cybersecurity research follows almost inevitably from the preceding architecture. Once AI is connected to knowledge, tools, memory, permissions, and external systems, every one of those connections becomes a potential attack surface. The conventional question "Is the model secure?" becomes insufficient.

The relevant architecture is

$$\text{Model} \to \text{Context} \to \text{Memory} \to \text{Knowledge} \to \text{Tools} \to \text{Permissions} \to \text{Communications} \to \text{Environment}.$$

An adversary may not need to compromise the model. It may be easier to manipulate what the model sees, what it remembers, which tools it trusts, which information it retrieves, which messages it receives from another agent, or which permissions surround it. This is why cybersecurity in the age of autonomous AI becomes inseparable from harnessing.

$$\boxed{\text{Model Security} \subset \text{Harness Security} \subset \text{System Security.}}$$

The author's cybersecurity work has emphasized precisely this expansion of the perimeter. Memory, context, permissions, communications, provenance, and delegated authority become security objects in their own right. Once AI systems can sustain actions over time, cybersecurity must also move beyond discrete event analysis. A sequence of individually legitimate actions can generate an unacceptable trajectory.

Suppose

$$a_1, a_2, a_3, a_4$$

are individually permissible actions. It does not follow that

$$(a_1, a_2, a_3, a_4)$$

constitutes a permissible trajectory. Intelligent adversarial systems may discover combinations of legitimate operations that create outcomes not anticipated by static security rules. The security architecture must therefore observe behavior through time.

"Cybersecurity moves from event security toward trajectory security."

The fifth harness adds authentication, segmentation, least privilege, behavioral monitoring, anomaly detection, containment, recovery, and incident reconstruction:

$$H_5 = H_4 + \{\text{authentication}, \text{least privilege}, \text{monitoring}, \text{containment}, \text{recovery}\}.$$

$$\boxed{H_5 = \text{Harness of Adversarial Resilience.}}$$

12. Stage VI: Spontaneous Collaboration and the Limits of Individual Harnesses

Research into spontaneous collaboration among autonomous agents pushes the argument one step further. Suppose agents A_1, A_2, A_3 are individually governed. Each has an identity, defined permissions, bounded tools, resource limits, and observable behavior. We might write

$$A_i \in H \qquad \forall i.$$

It is tempting to conclude that the system is therefore harnessed. That conclusion does not follow. If the agents communicate, adapt, observe one another, or respond to shared incentives, they may discover cooperative strategies that were never explicitly specified.

$$\boxed{A_1 \in H,\ A_2 \in H,\ A_3 \in H \not\Rightarrow (A_1, A_2, A_3) \in H.}$$

This is the transition from individual governance to systemic governance. The problem resembles phenomena familiar from economics and finance. Individually rational banks can collectively generate systemic instability. Individually rational traders can produce market dynamics that no participant intended. Local optimization does not guarantee desirable global outcomes.

The same possibility arises in artificial-agent populations. Let

$$G_t = (V, E_t)$$

represent the network of agents and their interactions. Governance of V, the nodes, is no longer sufficient. The institution must also observe E_t, the evolving relationships, and

$$\Phi(G_t),$$

the collective behavior generated by the network.

Harnessing therefore acquires a new dimension:

$$H_6 = H_5 + \{\text{interaction monitoring}, \text{communication governance}, \text{coalition detection}, \text{collective provenance}, \text{equilibr}$$

This connects the harnessing framework directly with the author's work on spontaneous cooperation and game-theoretical approaches to autonomous systems. The relevant question is not simply whether individual artificial agents behave rationally. It is whether their interactions generate equilibria that humans can identify, reconstruct, and govern.

$$\boxed{H_6 = \text{Harness of Emergent Interaction.}}$$

13. From Game Theory to Harnessing

The game-theoretical dimension is especially important because it provides a language for describing behavior that is neither random nor explicitly programmed. If autonomous agents respond strategically to one another, the system can be represented as a game

$$\mathcal{G} = \{N, S_i, U_i, I\},$$

where N represents agents, S_i their strategy sets, U_i their objectives or payoff structures, and I the information available to them.

Traditional governance might attempt to constrain S_i, the actions available to each agent. Emergent collaboration demonstrates that this may be insufficient. The equilibrium

$$s^* = (s_1^*, \dots, s_n^*)$$

is a property of the interaction among agents, not merely of any individual strategy set.

The harness may therefore need to govern not only actions but games. This is a significant conceptual extension. At the model level, we harness outputs. At the autonomous level, we harness actions and trajectories. At the multi-agent level, we may need to harness incentive structures, communication possibilities, coalition formation, and the topology of interaction itself.

The systemic question
What game have we allowed these agents to play? And, in the more difficult case, can the agents effectively create a game that we did not realize they were playing?

This is where provenance becomes systemic. We need not only action logs, but enough information to reconstruct the emergence of the interaction structure itself.

14. Harnessing Emergence Without Eliminating Emergence

The objective cannot simply be to prohibit agent-to-agent collaboration. Cooperative behavior may be one of the principal sources of value in multi-agent systems. Specialized agents may solve problems collectively that no individual agent can solve efficiently. The governance objective is therefore not to minimize interaction or emergence. It is to preserve beneficial emergence while identifying trajectories toward unacceptable collective behavior.

The relevant objective resembles

$$\max(\text{Useful Capability})$$

subject to acceptable bounds on risk, authority, observability, and reconstructability. The harness must therefore monitor not merely individual behavior but changes in the structure of interaction. Persistent communication among previously unrelated agents, unexpected specialization, reciprocal exchange, changes in task allocation, or the appearance of stable coalitions may become relevant signals.

This represents a transition from static rules toward behavioral governance. The institution no longer asks only whether a particular action is permitted. It asks whether a pattern of interaction is creating a new collective capability.

15. Stage VII: The Capstone — Complex Autonomous Systems

The ladder culminates in complex autonomous systems because this is where all the preceding themes converge. A self-driving vehicle provides a particularly clear example, but the concept extends far beyond transportation. Autonomous scientific laboratories, financial infrastructures, industrial control systems, logistics networks, cybersecurity architectures, and future institutional AI systems may all exhibit similar characteristics.

A complex intelligent system does not depend upon one universal model. Instead, it combines heterogeneous forms of artificial intelligence:

$$\mathcal{S} = \{M_1, M_2, \ldots, M_n, K, C, O, A\},$$

where M_i represent specialized models, K knowledge, C context, O optimization and orchestration mechanisms, and A action systems.

This returns us to the foundational observation of Stage I: AI is more than LLMs. But what was pedagogical at the beginning becomes architectural at the end. We begin the ladder by teaching that machine intelligence consists of a toolbox. We finish the ladder by constructing systems from that toolbox. The ladder therefore closes conceptually upon itself.

> ***"The capstone is not a bigger model. It is an integrated system of heterogeneous intelligence."***

16. Perception, Judgment, and Optimization as the Architecture of Intelligence

The complex system can be organized around three broad functions:

$$\boxed{\text{Perception} \rightarrow \text{Judgment} \rightarrow \text{Optimization.}}$$

These are not necessarily three separate models. They are functional categories within intelligent behavior. Perception concerns the construction of a usable representation of the environment. Computer vision, sensors, classification models, retrieval systems, anomaly detectors, and data-processing mechanisms may all contribute to this function.

Judgment concerns interpretation. It asks what the perceived state means, which information matters, what risks exist, and what objectives are relevant. Reasoning models, LLMs, probabilistic inference, causal analysis, institutional knowledge, and knowledge graphs may all contribute to judgment.

Optimization concerns choice. Given the perceived state, interpreted context, available actions, objectives, and constraints, what should the system do? Mathematical optimization, planning, search, reinforcement learning, scheduling, and control may contribute.

The full architecture is therefore

$$\boxed{\text{Environment} \rightarrow \text{Perception} \rightarrow \text{Judgment} \rightarrow \text{Optimization} \rightarrow \text{Action} \rightarrow \text{Environment.}}$$

Knowledge and context permeate the cycle. Memory allows it to operate through time. Cybersecurity protects it against manipulation. Provenance allows consequential behavior to be reconstructed. The harness surrounds the complete architecture.

17. The Self-Driving Vehicle as a Complete Harnessing Problem

Consider how this works in an autonomous vehicle. Sensors generate observations of the environment. Vision models identify lanes, vehicles, pedestrians, obstacles, and signs. Localization systems determine position. Prediction models estimate how surrounding actors are likely to move. A world model combines these observations with maps and contextual information. A judgment layer interprets the significance of the situation. A planner generates possible trajectories. An optimization mechanism evaluates them under safety, legal, comfort, time, and energy constraints. A control system translates the selected trajectory into steering, braking, and acceleration.

No single component is "the intelligence." The intelligence resides in the integrated architecture.

Now consider harnessing. The perception system must be harnessed because incorrect observations can contaminate every subsequent stage. The knowledge system must be harnessed because incorrect maps or rules can distort judgment. The reasoning layer must be harnessed because interpretation can fail. The optimization layer must be harnessed because a poorly specified objective can produce undesirable behavior even when perception and reasoning are correct. The control layer must be harnessed because a correct plan can still be executed incorrectly. Communications must be harnessed because external signals can be manipulated. The entire trajectory must be observable because system-level failure may not be attributable to a single component.

$$\boxed{\text{Component Safety} \not\Rightarrow \text{System Safety.}}$$

Likewise,

$$\boxed{\text{Component Governance} \not\Rightarrow \text{System Governance.}}$$

The capstone harness must therefore govern the architecture as a whole.

18. The Harness as a Control Architecture

At the complex-system level, the harness begins to resemble a control architecture. Let the intelligent system have state x_t, perceive observations y_t, construct judgments j_t, select optimized actions u_t, and affect the environment:

$$x_{t+1} = F(x_t, u_t, \epsilon_t),$$

where ϵ_t represents uncertainty.

The harness observes some representation of this process and imposes constraints:

$$u_t \in \mathcal{U}(H_t).$$

If the system is adaptive, however, the harness may also need to change:

$$H_{t+1} = G(H_t, B_t, R_t, E_t),$$

where B_t represents observed behavior, R_t estimated risk, and E_t the environment.

We therefore obtain two interacting adaptive processes:

$$\boxed{\text{Adaptive Intelligent System} \leftrightarrow \text{Adaptive Governance System.}}$$

This may represent one of the most important long-term directions for AI governance. Static rules will remain necessary, but they may not be sufficient for systems whose behavior, environment, and interaction topology change dynamically.

19. Provenance Becomes Reconstructability

Across the entire research program, provenance has repeatedly appeared as a central requirement. The harnessing framework reveals that provenance itself evolves. At Stage I, provenance means tracing an output to prompts and sources. At Stage II, it means reconstructing workflows. At Stage III, it means tracing machine knowledge to evidence. At Stage IV, it means reconstructing autonomous trajectories. At Stage V, it means identifying adversarial manipulation. At Stage VI, it means reconstructing the emergence of collective strategies. At Stage VII, it means understanding how heterogeneous components jointly produced a system-level outcome.

Thus,

$$\boxed{\begin{array}{c}\text{Provenance} \rightarrow \text{Process Lineage} \rightarrow \text{Knowledge Lineage}\\ \rightarrow \text{Behavioral Lineage} \rightarrow \text{System Reconstructability.}\end{array}}$$

This produces a general governance principle:

Consequential artificial intelligence must be sufficiently reconstructable to remain institutionally

Perfect interpretability of every neural activation is not required for this principle to be useful. What matters institutionally is the ability to reconstruct enough of the relevant context, evidence, authority, decisions, interactions, and state transitions to understand consequential behavior.

20. The Seven Harnesses

Table 2: The Seven Harnesses

Stage	Locus of Intelligence	Harness	Governance Object
I	Model	H_1	Inputs, outputs, identity and provenance
II	Architecture	H_2	Tools, workflows, memory and state
III	Knowledge system	H_3	Knowledge, evidence, lineage and memory
IV	Autonomous actor	H_4	Authority, objectives and trajectories
V	Adversarial system	H_5	Security, containment and resilience
VI	Agent network	H_6	Interaction, coalitions and emergence
VII	Complex system	H_7	Integrated cognition and action

The progression is cumulative:

$$H_7 = H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5 + \Delta H_6 + \Delta H_7.$$

The final harness therefore contains the institutional DNA established at the first stage: identity, provenance, validation, and accountability. What changes is the scale at which these principles must operate.

21. From Guardrails to Harnesses

The concept of a guardrail has become common in AI governance. Guardrails remain useful, but the metaphor becomes increasingly incomplete as systems acquire autonomy. A guardrail defines a boundary. A harness defines a relationship between something capable of movement and the entity responsible for controlling that movement.

This distinction captures the institutional problem more accurately. We do not want AI systems that cannot act. The economic value of autonomous intelligence derives precisely from its capacity to reason, explore, optimize, and execute useful work without requiring continuous human micromanagement.

The objective is therefore not

$$\min(\text{Autonomy}),$$

but rather

$$\max(\text{Useful Autonomy})$$

subject to

$$R \le R_{\max}, \qquad A \le A_{\max}, \qquad O \ge O_{\min}, \qquad Q \ge Q_{\min},$$

where R represents risk, A authority, O observability, and Q reconstructability.

Harnessing therefore does not mean suppressing intelligence. It means creating the architecture within which intelligence can be responsibly released.

> ***"A guardrail prevents departure. A harness permits movement while preserving connection, authority, and control."***

22. The Integration of the Research Program

Seen through the concept of harnessing, the different components of the author's research are not separate excursions into AI. They form a progression. The work on AI governance establishes identity, provenance, human responsibility, and institutional adoption. The work on agentic architectures demonstrates that intelligence becomes a property of orchestration rather than merely of the underlying model. The work on institutional knowledge shows that artificial cognition must be connected to structured organizational memory while preserving evidence and semantic provenance.

The work on autonomous systems introduces persistent agency and delegated authority. The cybersecurity research demonstrates that these new capabilities simultaneously create a radically enlarged attack surface. The research on spontaneous collaboration shows that individually governed agents can generate system-level behavior through strategic interaction. Finally, the work on complex intelligent architectures demonstrates where these developments converge: heterogeneous artificial intelligence operating through perception, judgment, optimization, and action.

Harnessing provides the missing transversal concept. It runs horizontally across a research program that otherwise appears vertical. The ladder describes increasing capability. The harness describes the corresponding increase in governance.

$$
\begin{array}{ccc}
\textbf{Capability} & & \textbf{Harnessing} \\
\text{Model} & \longleftrightarrow & \text{Output Governance} \\
\downarrow & & \downarrow \\
\text{Agentic Architecture} & \longleftrightarrow & \text{Process Governance} \\
\downarrow & & \downarrow \\
\text{Institutional Knowledge} & \longleftrightarrow & \text{Epistemic Governance} \\
\downarrow & & \downarrow \\
\text{Autonomous Agency} & \longleftrightarrow & \text{Authority Governance} \\
\downarrow & & \downarrow \\
\text{Cybersecurity} & \longleftrightarrow & \text{Adversarial Resilience} \\
\downarrow & & \downarrow \\
\text{Emergent Collaboration} & \longleftrightarrow & \text{Interaction Governance} \\
\downarrow & & \downarrow \\
\text{Complex Systems} & \longleftrightarrow & \text{System Governance.}
\end{array}
$$

This dual ladder is perhaps the clearest representation of the complete thesis. Capability and governance evolve together. The mistake would be to permit the left-hand side to climb the ladder while leaving the right-hand side behind.

23. Implications for Institutional Design

If the harnessing thesis is correct, the governance of artificial intelligence cannot remain concentrated in model-selection committees or acceptable-use policies. Those mechanisms remain necessary, but they govern only the lower levels of the ladder. As intelligence becomes more operational, governance must become architectural.

At the institutional level, this implies that AI architecture and governance architecture should be designed together. A system should not first be given tools, memory, institutional knowledge, persistence, and authority and only afterward subjected to a compliance review. Each new capability should have a corresponding control layer at the moment it is introduced. Identity should be designed into the agent architecture. Provenance should be designed into knowledge retrieval. Authority should be designed into tool permissions. Cybersecurity should be designed into communication and execution. Observability should be designed into autonomous trajectories. Systemic monitoring should be designed into multi-agent interaction.

This also implies a more demanding role for Boards and senior management. The relevant oversight question is no longer simply whether the organization has adopted an AI policy. It is whether the organization understands where intelligence resides within its increasingly complex architectures and whether it has established an adequate harness at the same level. An institution that deploys autonomous systems while retaining governance designed for conversational assistants creates a structural mismatch between capability and control.

The framework also has implications for regulators and public institutions. As artificial intelligence becomes embedded in financial, industrial, scientific, and critical infrastructures, model regulation alone may become insufficient. Regulatory analysis may need to examine harness design, knowledge provenance, delegation structures, interaction topologies, observability, and the ability to reconstruct consequential behavior. This is particularly relevant in sectors where individually rational actions can generate systemic effects.

24. Research Agenda

The harnessing framework suggests a broad research agenda. The first question is empirical: how much of observed agent capability is attributable to the underlying model and how much to the surrounding harness? This implies experimental comparisons in which model capability is held constant while context, memory, tools, orchestration, verification, or permissions vary.

A second question concerns institutional knowledge. How should knowledge graphs, retrieval systems, and persistent machine memory distinguish source evidence from generated inference? What mechanisms prevent machine-generated beliefs from recursively becoming accepted institutional facts? How should provenance travel when knowledge is transformed, summarized, or delegated across agents?

A third research problem concerns autonomy. Which variables best measure effective agency? How do tools, permissions, persistence, memory, and delegated objectives interact with model intelligence to produce operational capability? Can institutions define meaningful autonomy budgets that are more informative than simple tool-level access controls?

The cybersecurity dimension raises a fourth agenda. Traditional security evaluation tends to identify vulnerabilities or prohibited events. Autonomous systems require trajectory-level evaluation. How should security systems identify sequences of individually permitted actions that collectively produce harmful outcomes? How can behavioral drift be detected without eliminating useful adaptive behavior?

The fifth agenda concerns multi-agent emergence. What initial conditions, incentives, communication structures, or network topologies make spontaneous collaboration more or less likely? Can game-theoretical methods reconstruct the implicit games being played by artificial agents? Can a human observer reliably identify when a population of agents has moved from independent optimization into coalition formation or coordinated strategy?

Finally, complex autonomous systems require system-level evaluation. How should provenance be preserved across perception, judgment, prediction, optimization, and control? What does it mean to validate a system when no individual model can explain the complete behavior? How should an adaptive harness respond to changes in risk without creating a second opaque autonomous system around the first?

These questions reinforce the central claim of the paper. Harnessing is not merely an implementation detail. It is becoming an independent object of technical, institutional, and scientific inquiry.

25. Conclusion: Toward a Theory of Governable Intelligence

The history of institutional artificial intelligence may eventually be understood not simply as a history of increasingly powerful models, but as a history of progressively larger units of machine intelligence. We began with a model. We connected the model to tools and memory. We connected it to institutional knowledge. We allowed it to sustain objectives and delegated authority. We exposed it to an adversarial environment. We connected it to other artificial agents. We allowed artificial actors to interact. Ultimately, we began constructing complex systems in which many forms of intelligence cooperate to perceive, interpret, optimize, and act.

At every step, the relevant governance perimeter expanded. This is why harnessing provides such a useful organizing concept. Harnessing begins with provenance around a prompt, but it cannot end there. It expands around workflows, knowledge, memory, permissions, trajectories, communications, incentives, optimization, and eventually the complete intelligent system.

The central proposition of the research can therefore be stated as

As the locus of intelligence expands, the locus of harnessing must expand with it.

A second proposition follows:

Harnessing must occur at the same level at which intelligence becomes operational.

If intelligence resides in the model, harness the model. If intelligence resides in the architecture, harness the architecture. If intelligence depends upon institutional knowledge, harness the knowledge. If intelligence acts through time, harness the trajectory. If intelligence operates under adversarial conditions, harness the security perimeter. If intelligence emerges through interaction, harness the network and the game. If intelligence resides in a complex adaptive system, harness the system.

This reframes AI governance. The objective is no longer simply to make models safer. Nor is it merely to establish policies for responsible prompting or place restrictions around autonomous agents. The deeper institutional challenge is to build architectures capable of preserving identity, provenance, evidence, authority, observability, reconstructability, intervention, and accountability as machine intelligence becomes increasingly distributed and autonomous.

The capstone is therefore not the autonomous car itself. The autonomous vehicle merely makes the underlying principle unusually visible. The real capstone is the **complex intelligent system**: a system in which perception, knowledge, judgment, prediction, optimization, autonomous action, interaction, and feedback become integrated into a continuously operating architecture.

At that point, the question "Which AI model are we governing?" is no longer sufficient. The appropriate question becomes:

> **Where does intelligence reside in this system, and have we built a harness at that same level?**

That question connects the entire research program. It connects the first prompt to the autonomous agent, provenance to knowledge, knowledge to authority, authority to cybersecurity, cybersecurity to interaction, interaction to emergence, and all of them to the complex intelligent systems toward which artificial intelligence is increasingly moving.

Harnessing is the architecture through which the entire ladder remains governable.

Related Research and Institutional Context

This paper forms part of a broader research program examining how artificial intelligence can be converted from general technological capability into governed institutional value. The program follows the same progression developed in this paper: from responsible implementation, to agentic architectures, to institutional knowledge, to autonomous agency, to cybersecurity, and ultimately to the emergence and governance of complex interacting systems.

AI Governance 2026.
Establishes the general principles for implementing AI across corporate, governmental, and institutional processes. It addresses purpose, authority, accountability, evidence, human oversight, control design, and responsible deployment. Its central premise is that governance is not an obstacle to innovation, but the architecture through which AI capability becomes legitimate and durable institutional value.
`https://alexdibol.github.io/ai_summer_2026/`

Autonomous AI Systems.
Examines the transition from generative models that produce responses to agentic systems that can retain objectives, use memory, select tools, communicate, adapt to feedback, and execute multi-step processes. It provides the conceptual foundation for understanding the shift from artificial intelligence to artificial agency.
`https://alexdibol.github.io/ai_autonomous_systems/`

Second Brains and Knowledge Graphs.
Develops the institutional-knowledge dimension of the argument. It explains how governed second-brain architectures and knowledge graphs can preserve provenance, institutional memory, context, relationships, competing interpretations, and decision authority. The objective is not simply to give institutions access to more intelligence, but to ground that intelligence in trusted and accountable knowledge.
`https://alexdibol.github.io/ai_enhanced_intstitutional_ai/`

Cybersecurity in the Age of Agentic AI.
Explores how the cybersecurity perimeter changes when AI systems acquire memory, tools, permissions, communications capacity, adaptive persistence, and the ability to sustain action over time. Particular attention is given to integrity, provenance, delegated authority, containment, recovery, and the possibility that individually bounded agents may develop unexpected forms of cooperation.
`https://alexdibol.github.io/ai_cybersecurity/`

Autonomous Agency and Institutional Cyber Risk.
Brings these issues together from a Board and senior-management perspective. It considers how autonomous agency affects institutional authority, market and information integrity, strategic diagnosis, operational resilience, containment, remediation, and the conditions under which an autonomous system may responsibly remain—or return—to operation.
`https://alexdibol.github.io/biva_board/`

Selected References

The references below combine the author's research program with emerging work on agent harnesses, system scaling, autonomy, knowledge, cybersecurity, and complex systems. The purpose is not to imply that all of these literatures use the term *harnessing* in the same way. Rather, they establish the technical and intellectual foundations upon which the institutional extension proposed in this paper is built.

Harness Engineering and Agent Infrastructure

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OpenAI (2026). Harness Engineering: Leveraging Codex in an Agent-First World.
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Anthropic (2026). Scaling Managed Agents: Decoupling the Brain from the Hands.
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https://www.anthropic.com/engineering/managed-agents

Zhong, H. and Zhu, S. (2026). AI Harness Engineering: A Runtime Substrate for Foundation-Model Software Agents.
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Gu, S. (2026). From Model Scaling to System Scaling: Scaling the Harness in Agentic AI.
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Sutton, R. S. and Barto, A. G. Reinforcement Learning: An Introduction.
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Wooldridge, M. An Introduction to MultiAgent Systems.
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Shoham, Y. and Leyton-Brown, K. Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations.
Connects multi-agent architectures with game theory, strategic interaction, equilibria, mechanism design, and the formal analysis of autonomous-agent populations.

NIST (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0).
Provides a general institutional framework for governing AI risk through the functions of Govern, Map, Measure, and Manage, and remains a useful baseline against which the architectural extension proposed here can be understood.

NIST (2024). Cybersecurity Framework 2.0.
Provides a general framework for cybersecurity governance, identification, protection, detection, response, and recovery. The present paper extends these concepts to the changing perimeter of autonomous and agentic AI.

Saltzer, J. H. and Schroeder, M. D. (1975). The Protection of Information in Computer Systems.
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in this paper between component safety and system safety.

About the Author

Alejandro Reynoso is an Honorary Fellow and External Lecturer at Cambridge Judge Business School, University of Cambridge. His professional work spans public policy, financial-market institution building, investment banking, asset management, capital markets, and artificial intelligence. His current research examines how organizations can combine human judgment, institutional knowledge, generative AI, autonomous systems, cybersecurity, and governed knowledge architectures to create durable institutional intelligence.